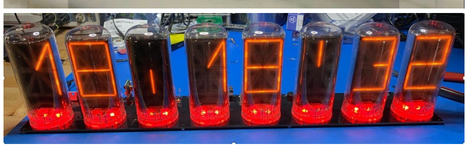
WordClock 1–4

Word Clock System

Operator’s Manual



April 2026

This manual describes the installation, configuration, and day-to-day operation of the WordClock 1, 2, 3, and 4 family.

Revision/Update Information:	Initial release
Clock Firmware:	V9.0.12

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Quick Start

Follow these steps to set up the WordClock for the first time. A full explanation of every option is in the numbered sections that follow.

1. Power on

Connect the clock to its 9–12V DC power adapter (1A or 2A minimum, depending on the model — see Section 2). Use a 5.5 × 2.1 mm center-positive barrel plug. On a brand-new clock, the display shows "AP" — this is Access Point configuration mode. (On a previously-used clock that already has WiFi credentials, skip to step 4 — to reconfigure WiFi, hold SW2 for 5 seconds during normal operation to enter AP mode.)

2. Connect to the clock's WiFi

From a phone or computer, open the WiFi settings and connect to the network named "WordClock-1-4" (or whatever model identifier matches the unit — for example, "WordClock-3-8" for an 8-digit VFD). No password is required.

3. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup form loads. At minimum, navigate through the pages and:

- On the Setup page, select the home WiFi network from the SSID list and enter the password.
- Confirm the UTC offset for the local region (US Eastern is 5 hours Minus by default).
- On the Libraries page, confirm Library 1 is set (the default English library is required).

Press Save at the bottom of the form to store the selections, then press Start. The clock saves the settings, reboots, and begins normal operation.

4. Wait for time to appear

During boot, the display briefly scrolls "Wi-Fi" while connecting to the network, then "NTP" while fetching time from the internet. The display also scrolls each enabled library name and an "Index" or "No Idx" status for each. Time and rotating words appear within a few seconds of NTP sync.

5. Daily use

To do this	Do this
Wake the display from a quiet period	Click SW1 once
See the clock's IP address	Double-click SW1 — the address scrolls across the display
Cycle the expletive level (Off/Mild/Medium/Strong)	Click SW2 once
Adjust most settings live	Browse to http://<ip-address>/ from any device on the same WiFi
Send a custom scrolling message to the display	Use the Display Message field on the live web page
Install new firmware over WiFi	Hold SW1 for 10 seconds

6. Recommended next steps

- Visit the live web interface to explore backlight effects, colon patterns, word transitions, and other display options.
- Configure Period Overrides on the live page to dim or turn off the display during sleeping hours.
- If a BME280 weather sensor is installed, enable inside temperature and humidity on the Info Display page. To enable outside weather, register for a free Visual Crossing API key at visualcrossing.com and enter the key, postal code, and country on the Info Display page.

1. Overview

WordClock uses alphanumeric LED displays, VFD displays, or B7971 Nixie tubes to present the time, date, and rotating word libraries in a distinctive text-based format. Models are available with 6, 8, or 16 characters depending on display type.

The standard software includes an English word library, optional expletive files with selectable rudeness levels, announcements by month and day, and support for additional user-supplied libraries and languages. Words and phrases are stored on a removable SD card as plain-text files that can be edited with any text editor.

WordClock can synchronize time from a Wi-Fi NTP server or from an optional GPS receiver. Additional features include three independent display override periods, RGB backlight effects, inside and outside weather display, a live browser-based settings page, custom scrolling messages, and over-the-air firmware updates.

All settings are stored in non-volatile memory (NVS) and survive power cycles. Configuration is performed through a Wi-Fi access point setup page or a live web interface accessible from any device on the same network.

Note: Some included libraries — particularly the British slang collections and BadW libraries — contain language that some users consider offensive. Whilst that is somewhat the point, if young children or sensitive guests are present, the expletive level can be set to Off via SW2 or the live web page.

2. Models

WordClock is sold in several display variants. The hardware variant is set at the factory and identified by a code stored in the clock's flash memory; it cannot be changed by the user. The Access Point name and mDNS hostname both include the model identifier.

Model	Digits	Display Type	Notes
WordClock-1-6	6	LED (alphanumeric)	0.8" digits, single or dual LED ring backlight
WordClock-1-8	8	LED (alphanumeric)	0.8" digits
WordClock-1-16	16	LED (alphanumeric)	0.8" digits, dual I²C bus
WordClock-2-6	6	LED (alphanumeric)	2.3" digits
WordClock-2-8	8	LED (alphanumeric)	2.3" digits
WordClock-2-16	16	LED (alphanumeric)	2.3" digits, dual I²C bus
WordClock-3-6	6	VFD (IV4 / IV17)	Vacuum-fluorescent tubes
WordClock-3-8	8	VFD (IV4 / IV17)	Vacuum-fluorescent tubes
WordClock-3-16	16	VFD (IV4 / IV17)	Vacuum-fluorescent tubes, dual I²C bus
WordClock-4-6	6	Nixie (B7971)	Alphanumeric Nixie tubes, HV5530 driver
WordClock-4-8	8	Nixie (B7971)	Alphanumeric Nixie tubes, HV5530 driver

Power supply: 9–12V DC, center positive, 5.5 × 2.1 mm barrel plug. The 6-digit LED models are typically supplied with a 1A adapter; 8- and 16-digit LED, VFD, and Nixie models are supplied with 2A. Use a quality regulated supply.

If the display shows "BAD" at power-on and never starts, the model code stored in flash is missing or invalid. Contact the distributor.

3. Display

Depending on the model, the clock has 6, 8, or 16 alphanumeric character positions. Each position is a 14- or 16-segment alphanumeric element capable of rendering ASCII letters, digits, and a number of special characters. RGB backlight LEDs sit behind each character and can run a variety of patterns (see Section 9).

A status indicator behind the leftmost character blinks briefly when Wi-Fi or NTP/GPS connectivity is lost. This blink can be suppressed via the live or setup form if desired.

On models that include colon segments between digit pairs, the colons can be configured to flash, scan, or stay solid (see Section 7, Colon Display).

4. Buttons

The clock has two buttons: SW1 (left) and SW2 (right). Their behavior differs between normal operation and power-on.

SW1 — During Normal Operation

Action	Function
Single click	Wake the display from a quiet period and start the temporary button-override timer (15 s, 1 min, or 5 min — see Section 8)
Double-click	Scroll the current IP address (or mDNS name) across the display when Wi-Fi is connected
Hold 10 seconds	Trigger an over-the-air (OTA) firmware update — see Section 15

SW2 — During Normal Operation

Action	Function
Single click	Cycle the expletive level: Off → Mild → Medium → Strong → Off. The level name briefly appears on the display.
Triple-click	Show the raw light-sensor reading for 30 seconds (calibration mode)
Hold 5 seconds	Reboot into Access Point configuration mode while preserving the current settings

Button Override

Pressing SW1 during a quiet period (a configured override) temporarily wakes the display and forces it to show the time only. The wake lasts for the configured Button Override duration: 15 seconds, 1 minute, or 5 minutes (default 15 seconds). After the timer expires, the display returns to its quiet-period state.

5. Power-On Functions

Special functions are available by holding buttons while applying power. These are read before the clock initializes its normal button handling. Apply power to the clock with the indicated buttons held; release once the clock acknowledges the action.

Buttons Held	Function
SW1 only	Rebuild word library indexes from the SD card. The display shows "Index" while running. See Section 12 for details on per-library rebuild and full rebuild options.
SW2 only	Burn-in mode. All segments stay on at full brightness until any button is pressed. Useful for exercising IV4 / IV17 VFD tubes on WordClock-3 models. Not available on Nixie variants.
SW1 + SW2	Factory reset. Clears all stored settings (Wi-Fi credentials, display preferences, period overrides, weather, etc.) and reboots into Access Point mode. The hardware model code is preserved.

6. AP Configuration Mode

Entering AP Mode

- Hold SW2 for 5 seconds during normal operation, OR
- Apply power with SW2 held down (use the long-press method above for systems already configured), OR
- Power on a brand-new clock — it boots directly into AP mode

In AP mode the display shows "AP" continuously.

Connecting

1. Connect a phone or computer to the Wi-Fi network broadcast by the clock — by default "WordClock-1-4" (or "WordClock-3-8", etc., matching the model). The AP is open (no password).

2. Open a web browser and go to 192.168.4.1.

3. Step through the pages of the setup form. Press Save when finished, then Start to apply the settings and reboot.

The setup form is split across multiple pages — Setup (network and time), Libraries, Display, Overrides, and Info Display — to keep the interface manageable. Use the page buttons on the home page to navigate.

Setup Page — Network

Setting	Description
Access Point Name	Wi-Fi network name broadcast by the clock when in AP mode (default: WordClock-X-X matching the model)
SSID	Select the home Wi-Fi network from the scan list. SSIDs containing spaces may fail to connect.
SSID Override	Manually type an SSID if it does not appear in the scan list (for hidden networks). Overrides the dropdown selection.
Network Password	Wi-Fi password (max 25 characters; trailing spaces are not supported).
Live Form Password	Optional password for the live web interface. Leave blank for no password.
Primary NTP Server	Primary internet time server (default:

	time.cloudflare.com)
Alternate NTP Server	Backup time server, used if the primary becomes unreachable for more than 30 minutes (default: time.google.com)
Live Form	Enables or disables the live web interface during normal operation (default: Enabled)

Note on NTP servers: many home routers offer a built-in NTP server. Pointing all home devices to the router (e.g. 192.168.1.1) avoids excessive load on public servers and complies with the NTP Pool Project terms of service, which ask that SNTP clients query no more than once every 30 minutes.

Setup Page — Time and Time Zone

Setting	Options	Default
Time Source	Wi-Fi/NTP, GPS, Off	Wi-Fi/NTP
GPS Baud Rate	4800, 9600, 19200, 38400, 57600	9600
Time Display	12 Hour, 24 Hour, US English Verbose	12 Hour
UTC Offset	HH:MM (supports half-hour and 45-minute regions)	5:00
UTC Offset Direction	Minus, Plus	Minus

The Verbose time format spells the time in words — for example, "FIVE MINUTES AFTER 3" or "MIDNIGHT" — and works best on 8- and 16-digit displays.

Setup Page — Daylight Saving Time

Setting	Options	Default
DST Start Month	Jan–Dec	March
DST Start Week	1–5	2
DST Start Day	Sun–Sat	Sunday
DST Start Hour	0–23	2
DST Offset	0 hr, 1 hr	1 hr
DST Direction	Plus, Minus	Plus

Setup Page — Standard Time

Setting	Options	Default
ST Start Month	Jan–Dec	November
ST Start Week	1–5	1
ST Start Day	Sun–Sat	Sunday
ST Start Hour	0–23	2

Default DST and Standard Time settings are configured for US Eastern Time.

Libraries Page

Up to five word libraries can be active at once. Library 1 is required (it provides the system message file used for all status text and is the default content source). Libraries 2–5 are optional and only used when the corresponding Use Library percentage is non-zero. Each library is a folder on the SD card with files sharing the folder's base name.

Setting	Options	Default
Library 1	Any folder on the SD card	English
Library 2 / 3 / 4 / 5	Any folder on the SD card, or none	none
Use Library 2 / 3 / 4 / 5	Off, 5%, 20%	Off
Time and Date Language	Always Library 1, Current Library	Always Library 1

"Always Library 1" forces the time and date to use the English (or Library 1) font and number labels even when a foreign-language library is currently displaying words. "Current Library" uses the active library's font for time and date.

Display Page

Setting	Options	Default
Display Brightness	0–9	9
Clock Mode	Off, Every 15s, Every 30s, Every 60s, Always On	Every 30s
Colon Display	9 patterns (off, on, scan, flash, etc.)	Pattern 4
Word Change Effect	Off, Slide Up, Slide Down, Slide Left, Slide Right, Energize, Random	Energize
Time Change Effect	Off, Slide Up, Slide Down, Slide Left, Slide Right, Random	Slide Right
Scroll All Words	Yes, No	No
Show Index Status at Start	On, Off	On
Show Announcements	Off, Every 60 s, Every 30 s	Every 30 s
Announcement Override	Off, On	Off
Backlight Effect	14 effects + Cycle (see Section 9)	Random Multi-Color Wave
Backlight Brightness	0–9	9
Expletive Default	Off, Mild, Medium, Strong	Off
Expletive Mix	5%, 20%, 50%	5%

Clock Mode controls how often the time replaces the rotating word display. "Always On" disables word rotation entirely. "Scroll All Words" forces every word — even those that fit the display — to scroll horizontally; otherwise short words are shown statically.

Overrides Page

See Section 8 for full details. Three independent quiet/dim periods can be defined here, plus the Button Override duration (15 s / 1 min / 5 min) used after a SW1 click.

Info Display Page

Setting	Options	Default
Info Display Start Second	0–59	4
Date Display	MM/DD/YY, DD/MM/YY, YY/MM/DD, US English Verbose, Off	MM/DD/YY
Light Sensor	Disabled, Enabled	Disabled
Min Display Brightness	0–9	3
Min Backlight Brightness	0–9	3
Motion Sensor Timeout	Off, 15 s, 1 min, 5 min, 10 min, 15 min, 30 min, 1 hr	Off

See Section 11 for the date / weather / announcement display sequence and Section 13 for sensor details.

Info Display Page — Weather

Visible only when a Visual Crossing API key is configured or a BME280 inside sensor is detected.

Setting	Options	Default
Weather Key	Visual Crossing API key (text)	blank
Postal Code or Lat/Lon	Location for weather lookups	blank
Country Code	Two-character country code (e.g. US, GB, AU)	US
Inside Weather Display	Off, Temperature, Temperature + Humidity	Off
Outside Weather Display	Off, T, T+H, T+H+P, T+H+P+W	Off
Current Conditions	Off, On	Off
Scale	Imperial (°F, mph, inHg), Metric (°C, km/h, mb), Kelvin	Imperial
Inside Temperature Offset	−99 to +99 (0.1° increments)	0
Labels	Off, Short, Long	Off
Display Duration	5 seconds, 10 seconds	5 seconds

OTA Update

The Firmware URL field on the Setup page configures where the clock fetches new firmware from when an OTA update is triggered. The default value (https://mitchsf.github.io/Firmware/) points to the manufacturer's firmware server. This field cannot be left blank; if cleared, it reverts to the default. See Section 15 for the update procedure.

7. Live Settings Web Interface

Once the clock is connected to a Wi-Fi network, the live settings page is reachable from any device on the same network. Most settings can be adjusted live without rebooting the clock — the change takes effect the moment the control is released.

Access: Browse to http://<clock-ip-address>/. To find the IP address, double-click SW1 — the address scrolls across the display.

Using a Name Instead of an IP Address (mDNS)

On many home networks the clock can be reached by name instead of by IP address. Browse to http://wordclock-1-4.local/ (or wordclock-3-8.local, wordclock-2-16.local, etc., matching the model) and the page should load. The hostname is derived from the Access Point Name set during configuration: lowercased, with any spaces or non-alphanumeric characters converted to hyphens. For example, an AP Name of "WordClock-1-4" becomes wordclock-1-4.local; "Kitchen Clock" becomes kitchen-clock.local. This makes it easy to give each clock in a multi-clock household a memorable address.

mDNS (also known as Bonjour or zero-configuration networking) does not work on every network. Some routers — particularly older models or routers with multicast filtering enabled — block the discovery messages mDNS depends on, in which case the .local address will not resolve. macOS, iOS, and Windows 10/11 generally work out of the box; some Android versions and Linux distributions require additional setup. If the .local address does not load, fall back to the IP address (double-click SW1 to display it).

Authentication

If a Live Form Password was set during AP configuration, the page prompts for it before any settings are shown. The username is left blank; enter the configured password. Clear the field in AP mode to disable authentication.

Display Page Controls

All changes take effect immediately. Use the Save to Memory button on each page to persist changes across power cycles; live changes alone are lost on reboot. The form is split across multiple pages with a home menu; the address bar shows which page is active.

- Display Brightness (0–9)

- Backlight Brightness (0–9)
- Backlight Effect (all 14 effects, see Section 9)
- Colon Display (9 patterns)
- Clock Mode (Off, Every 15 s, Every 30 s, Every 60 s, Always On)
- Word Change Effect (Off, four slide directions, Energize, Random)
- Time Change Effect (Off, four slide directions, Random)
- Show Announcements (Off, Every 60 s, Every 30 s)
- Announcement Override (force announcements during Always On)
- Expletive Default level and mix percentage

Overrides Page Controls

- Override Mode toggle (Enabled / Disabled) — temporarily suspends all three override periods without deleting them. This setting is not saved; on reboot it returns to Enabled.
- All settings for Override 1, 2, and 3 — start time, end time, days, brightness, backlight effect and brightness, clock mode, colon display
- Button Override Duration (15 s, 1 min, 5 min)

Info Display Page Controls

- Info Display Start Second (0–59)
- Date Display format
- Motion Sensor Timeout
- Light Sensor enable, Min Display Brightness, Min Backlight Brightness
- Inside / Outside weather display modes, scale, temperature offset, labels, duration

Custom Scrolling Message

The home page of the live web interface includes a Display Message text field (up to 64 characters). Type a message and press Send. The message scrolls across the display once at the next opportunity, then the clock returns to its normal display sequence. This is useful for greetings, reminders, or quick announcements visible to anyone in the room.

Override Mode Toggle

Override Mode is a "master switch" for all three quiet-period schedules. When set to Disabled, the clock ignores every period and runs at standard brightness and effects regardless of the time of day. This is helpful when guests are visiting and the configured night-time dim is not wanted. The toggle lives on the Overrides page and is always visible there. The setting is intentionally not saved to memory — on reboot, Override Mode returns to Enabled so the configured schedules resume automatically.

8. Period Overrides

Three independent override periods can each schedule a different display configuration for specific times of day and days of the week. The most common use is to dim or turn the display off during sleeping hours.

Setting	Description
Start Time	When the period begins, in 24-hour HHMM format (e.g. 2300 = 11:00 PM)
End Time	When the period ends, in 24-hour HHMM format (e.g. 0700 = 7:00 AM)
Days	All days, Weekdays only, or Weekends only
Display Brightness	0–9 (0 = display off; on Nixie models the high-voltage supply is also disabled)
Backlight Effect	Any backlight effect, including Off
Backlight Brightness	0–9
Clock Mode	Off, Every 15 s, Every 30 s, Every 60 s, Always On
Colon Display	Any colon pattern, including Off

Setting Start Time equal to End Time disables the period. Periods may wrap across midnight (e.g. 2200 to 0700).

Priority: when two or more periods overlap, the lowest-numbered period wins (Period 1 > Period 2 > Period 3).

Override Mode (live web page) suspends all three periods without erasing them. When Override Mode is Disabled the clock runs at standard settings regardless of the time of day.

Button Override: a single click of SW1 during an active period temporarily wakes the display and forces Clock Mode 4 (always on, time only) for the configured Button Override duration (15 seconds, 1 minute, or 5 minutes). When the duration expires, the period's settings resume.

9. Backlight (LED) Effects

A row of RGB LEDs sits behind the alphanumeric characters and provides a colored backlight. The number of LEDs depends on the model — some 6-digit clocks use 6 LEDs, others use 12 or 24; 8-digit clocks use 8, 16, or 32; the largest 16-digit clocks use up to 64. Effects scale automatically to the LED count.

#	Effect	Description
0	Off	All LEDs dark
1	USA Wave	Red, white, and blue cycling with sine-wave brightness modulation
2	Israel Wave	Blue and white alternating wave
3	Random Big Color Wave	Single random color (8-bit RGB) with sine-wave fade
4	Random Every Color Wave	Single random full-spectrum color with sine-wave fade
5	Random Multi-Color Wave	Different random color per LED with sine-wave fade (default)
6	Twinkle	Random colors per LED, fading independently
7	Red Fade	Red sine-wave fade
8	Green Fade	Green sine-wave fade
9	Blue Fade	Blue sine-wave fade
10	Solid Red	Steady red, no animation
11	Solid Green	Steady green, no animation
12	Solid Blue	Steady blue, no animation
13	Cycle 1–9	Rotates through effects 1–9, switching every 10 seconds at the top of each minute

Backlight Brightness (0–9) is gamma-corrected so that small numbers correspond to dim, evenly-spaced steps and large numbers ramp up to maximum. Setting Brightness to 0 turns the LEDs off without changing the selected Effect.

Status indication: when the time source (Wi-Fi/NTP or GPS) is lost, the leftmost LED briefly turns off and back on roughly every 1.5 seconds. This indicator can be temporarily blanked by the configured period override.

10. Word and Time Change Effects

Two independent transition effects animate how new content appears on the display.

Word Change Effect

Applied each time a new word from the active library replaces the previous word.

Effect	Description
Off	Instant replacement, no animation
Slide Up	New word slides in from below
Slide Down	New word slides in from above
Slide Left	New word slides in from the right
Slide Right	New word slides in from the left
Energize	Per-segment "energize" sweep — looks particularly good on VFD models
Random	Picks a random slide direction each time

Time Change Effect

Applied each time the displayed time updates (controlled by Clock Mode — see Section 7).

Effect	Description
Off	Instant replacement, no animation
Slide Up	New time slides in from below
Slide Down	New time slides in from above
Slide Left	New time slides in from the right
Slide Right	New time slides in from the left (default)
Random	Picks a random slide direction each time

11. Info Display Cycle

The clock periodically interrupts the rotating word display to show date, weather, and announcement information. Most info shown depends on the configured Date Display, weather, and announcement settings.

Information Shown (if enabled)

Date: Displayed as MM/DD/YY, DD/MM/YY, YY/MM/DD, or US English Verbose ("April 29, 2026"). Set Date Display to Off to skip the date.

Inside Weather: Read from a BME280 sensor on the I²C bus (auto-detected at addresses 0x76 and 0x77). Temperature alone, or temperature with humidity. The Inside Temperature Offset can be used to correct sensor calibration.

Outside Weather: Fetched from Visual Crossing (https://visualcrossing.com — a free API key is required). Up to five elements can be displayed: temperature, humidity, pressure, wind, and conditions. Wind shows as both speed and 16-point compass direction (N, NNE, NE, ENE, E, ...).

Current Conditions: A short text description (e.g. "PARTLY CLOUDY") appended to the outside weather display when enabled.

Announcements: Day-of-month or month-and-day messages from the Library 1 announcements file (see Section 12). Shown every 30 or 60 seconds when enabled, or on demand at midnight.

Display Timing

Info Display Start Second (0–59) controls when in each minute the date / weather / announcement cycle begins. Display Duration (5 s or 10 s) controls how long each weather element is held on the display before transitioning to the next.

Labels can be turned Off, Short ("T-72.4F"), or Long ("TEMPERATURE 72.4F") to fit different display widths.

Weather Update Schedule

When outside weather is enabled, the clock fetches a fresh forecast every 15 minutes. If the fetch fails, it retries after 5 minutes. Data older than 1 hour is treated as stale and the previous values are cleared until a successful fetch.

12. Word Libraries

Words, fonts, system messages, expletives, and announcements are all stored on a removable SD card. Each library is a folder at the root of the card, and the files inside share the same base name as the folder. Up to five libraries can be active at once; the SD card may contain up to 25 library folders to choose from.

Folder Structure

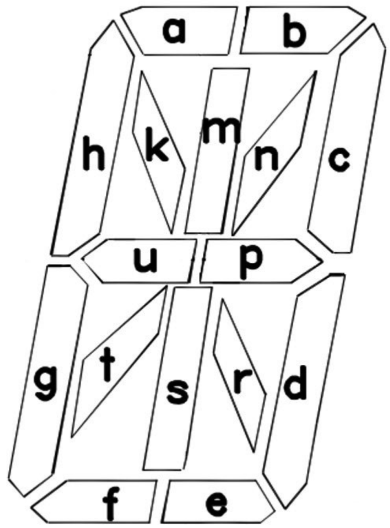
Each library folder contains a font file and a word file at minimum. Optional files add expletives, announcements, system messages, and a search index. For example, an English library would contain:

- /English/English.fnt — character bitmaps (required)
- /English/English.wds — word and phrase list (required)
- /English/English.idx — search index (optional, auto-generated)
- /English/English.msg — system messages (required for Library 1 only)
- /English/English.xpl — expletive list with rudeness levels (optional)
- /English/English.ano — month/day announcements (optional, Library 1 only)

Library names may be up to 8 characters long and cannot contain spaces. Folders beginning with a dot (.) and the special folder EXTRAS are skipped.

Fonts (.fnt)

Plain-text files. Each line begins with 16 zeros and ones describing the segment states for one character, in the order: u t s r p n m k h g f e d c b a. ASCII characters from 32 to 255 may be defined within the limits of the display hardware. Comments after the 16-character segment definition are ignored.



The diagram above shows the segment letter map for one alphanumeric character — each of the sixteen segments corresponds to one bit in the font definition.

Words (.wds)

Plain-text. Each line is a single word or phrase up to 64 characters. Two short words can share one line by separating them with a pipe (|). The first line is treated as a header comment and is never displayed.

Messages (.msg)

Required only in Library 1. The default message file supplies the strings shown for system events (Wi-Fi, GPS, NTP, AP, expletive level names, weather, etc.). Required entries must appear in the expected order — see the supplied default file for the canonical layout.

Expletives (.xpl)

Optional. Each line contains a word or phrase, a pipe (|), and a level from 1 to 3. Higher levels include all lower levels (level 3 = Strong includes Mild and Medium content). Up to 125 entries per file. The first line is treated as a header comment.

Announcements (.ano)

Optional, Library 1 only. Each line begins with day and month in DD-MM format followed by the announcement text (up to display-width characters). A month value of 00 fires the announcement on that day of every month.

Indexes (.idx)

A binary-style index that lets short and long entries be selected with equal probability. Whenever a word file is edited, its index must be rebuilt. There are three ways to do this:

- To rebuild every library, hold SW1 while applying power. Release once "Index" appears. The clock indexes each library in turn before continuing the boot sequence.
- To rebuild a single library, watch the boot screen — each enabled library name briefly scrolls. Press and hold SW1 while the desired library name is shown, and only that library will be reindexed.
- Indexes are automatically generated on first use if missing — the clock displays "No Idx" briefly during the boot sequence to indicate which libraries lack an index.

Library 1 is required at all times. Removing it (or replacing it without a message file) prevents the clock from booting normally.

13. Sensors

Light Sensor

When enabled (Light Sensor = Enabled), the clock continuously samples ambient light and dims both the display and backlight in dim rooms. The Min Display Brightness and Min Backlight Brightness settings prevent the display from going completely dark. With the sensor disabled, the configured brightness values apply at all times.

Calibration: triple-click SW2 to enter calibration mode. The raw sensor reading (a small number from 0 to 15) is shown on the display for 30 seconds. Use this to confirm the sensor is responding and to set the Min Brightness values appropriately for the room.

Motion Sensor (PIR)

When the optional PIR motion sensor is installed and a non-Off timeout is configured, the display blanks after the timeout has elapsed with no detected motion. When motion is detected, the display fades back in to the configured brightness; when motion stops and the timeout expires, the display fades out again. Available timeouts: 15 s, 1 min, 5 min, 10 min, 15 min, 30 min, and 1 hour.

A 30-second warm-up period applies after every power-on so that the sensor has time to stabilize.

BME280 Weather Sensor

If a four-pin BME280 module is connected to the I²C bus, inside temperature, humidity, and barometric pressure are auto-detected at startup (addresses 0x76 and 0x77). When detected, the Inside Weather Display options become active in the setup and live forms. The Inside Temperature Offset (−99 to +99 in 0.1° steps) can be used to compensate for sensor self-heating.

14. Time Source

Wi-Fi / NTP

The default time source. The clock fetches time from a Network Time Protocol (NTP) server roughly every four hours, with random jitter to avoid synchronized polling. If the primary server (default time.cloudflare.com) is unreachable for more than 30 minutes, the alternate server (default time.google.com) takes over automatically. The clock falls back to the primary as soon as it becomes responsive.

NTP requires Internet access. Without an Internet connection the clock continues to run on the ESP32's internal oscillator but slowly drifts; use a router-hosted NTP server for offline networks.

GPS

An optional GPS receiver can be connected to the hardware serial port (RX on GPIO 12, GND, and 3.3V or 5V depending on the receiver). When Time Source is set to GPS, the clock derives time directly from satellite fixes and does not require Wi-Fi. The GPS Baud Rate setting must match the receiver — most modern receivers use 9600 (the default).

A clear sky view is needed for the initial fix. The display shows "GPS" while waiting for a valid time message.

Off

Disables time synchronization entirely. The clock runs purely from the internal oscillator, which drifts on the order of seconds per day. Useful for table-top demonstrations or when no time source is available.

15. OTA Firmware Update

The clock can update its own firmware over Wi-Fi. To trigger an update, hold SW1 for 10 seconds during normal operation. The clock checks the configured firmware server, downloads any newer version, and reboots automatically.

Display Feedback

OTA status messages appear on the full display as scrolling text or short codes. The progression is:

Display	Meaning
UPDATE	Update started. The clock is connecting to Wi-Fi and the firmware server.
LATEST	The installed firmware already matches the version on the server. The clock displays the version briefly, then returns to normal operation.
(progress)	A new firmware binary is being downloaded and written to flash.
OK	Update succeeded. The clock reboots automatically after a short delay.

Error Codes

If the update cannot complete, an error code is shown briefly and the clock returns to normal operation:

Code	Meaning
ERROR1	No Wi-Fi connection or no firmware URL configured.
ERROR2	The VERSION file on the firmware server could not be retrieved.
ERROR3	The VERSION file is malformed (missing or invalid version string).
ERROR4	The firmware binary failed to download or could not be written to flash.

Version Behavior

Major version change (e.g. 8.x.x to 9.x.x): all settings are reset to factory defaults after the update. This guarantees clean migration when the configuration format changes between major releases.

Minor or patch change (e.g. 9.0.11 to 9.0.12): all settings are preserved.

Requirements

A working Wi-Fi connection and a valid Firmware URL are required. The default URL points to the manufacturer's firmware server (https://mitchsf.github.io/Firmware/) and should be left at the default unless directed otherwise.

Manual Reflash

In rare situations — for example, if the clock will not boot far enough to perform an OTA update — firmware can also be loaded over USB using Espressif's Flash Download Tool. Detailed manual-reflash instructions are available from the distributor on request.

16. Factory Reset

To reset all settings to factory defaults:

1. Remove power from the clock.
2. Press and hold both SW1 and SW2.
3. Apply power while continuing to hold both buttons.
4. Continue holding until the display clears (about 5 seconds).
5. Release both buttons. The clock restarts in AP configuration mode.

All Wi-Fi credentials, display preferences, period overrides, weather settings, and library selections are erased. The hardware model code (which identifies the unit as a WordClock-1-4, WordClock-3-8, etc.) is preserved.

A factory reset is also performed automatically when an OTA update changes the major version number (see Section 15).

17. Display Messages

In addition to time, words, weather, and announcements, the clock displays a number of short status messages. The most common are listed here.

Display	Meaning	Operator Action
AP	Access Point configuration mode — waiting for setup.	Connect to the WordClock-X-X Wi-Fi and configure at 192.168.4.1.
Wi-Fi	Connecting to a Wi-Fi network.	Wait for connection. If it does not complete, verify SSID and password in AP mode.
NTP	Connecting to the NTP time server.	Wait for time sync. Check internet connectivity if it stalls.
GPS	Waiting for a GPS fix.	Ensure the receiver has a clear view of the sky.
WX	Fetching weather data.	Wait for the fetch to complete or verify API key and location.
Index	A search index exists for the displayed library (boot only).	No action required.
No Idx	No search index exists for the displayed library.	Rebuild the index if the library has been edited (see Section 12).
SDcard	The SD card cannot be mounted or is not present.	Reseat or replace the card. Format as FAT32 if needed.
Font X	The font file for Library X is missing.	Install the missing .fnt file in the affected library folder.
Word X	The word file for Library X is missing.	Install the missing .wds file in the affected library folder.
Mesg 1	The message file for Library 1 is missing.	Add the required .msg file to Library 1.
BAD	Software licensing or hardware-code violation.	Contact the distributor.
UPDATE / OK / ERROR1–4	OTA firmware update progress (see Section 15).	See the OTA section for the appropriate response.

Status indicator: the leftmost backlight LED briefly turns off and back on every 1.5 seconds whenever the time source is unavailable. This blink can be hidden during a configured override period if desired.

18. Troubleshooting

Symptom	Possible Cause	Solution
Display is blank after power on	Waiting for Wi-Fi association or NTP time sync	The display shows "Wi-Fi" while connecting and "NTP" while syncing. If neither completes within ~2 minutes the software watchdog reboots the clock and tries again.
Display shows steady "AP"	Clock is in AP configuration mode	Connect to the WordClock-X-X Wi-Fi and configure at 192.168.4.1.
Display shows steady "BAD" and never starts	Missing or invalid hardware model code	Contact the distributor; the unit needs re-provisioning.
Display shows "SDcard"	SD card not present, unreadable, or wrong format	Reseat the card or reformat as FAT32 with at least Library 1 installed.
Display is dim during the day	Light sensor active and ambient light is low	Triple-click SW2 to view the raw sensor value. Increase Min Display Brightness or disable the sensor.
Status LED blinks briefly off	Wi-Fi or NTP/GPS connection lost	The clock is reconnecting automatically. If the blink persists, check the router or re-enter AP mode.
Wi-Fi will not connect at boot	Wrong credentials or out of range	Hold SW2 for 5 seconds to enter AP mode and reconfigure Wi-Fi.
Time is wrong by a whole hour	Incorrect DST or Standard Time configuration	Check DST/ST start month, week, day, and offset in AP setup.
Time is briefly wrong, then corrects	Primary NTP server slow; alternate took over	Normal behavior. The clock falls back to the alternate NTP server after the primary is stale for 30 minutes.
Words look funny / characters missing	Font file for the active library is corrupt	Reinstall the .fnt file from the distributor's library package.
Library does not appear in setup form	Library folder name has spaces or is too long	Rename to ≤ 8 characters with no spaces and reseat the SD card.
New words from edited library are not appearing	Word file edited but index not rebuilt	Hold SW1 at power-on (or while the library name scrolls) to rebuild the index.
Weather "WX" appears repeatedly without succeeding	Wrong API key, postal code, or country code	Re-check the Visual Crossing key and the location format in AP setup.
Inside weather options not visible in form	BME280 sensor not detected on the I²C bus	Reseat the sensor; verify wiring; confirm address 0x76

		or 0x77.
OTA shows ERROR1	No Wi-Fi or missing Firmware URL	Reconnect to Wi-Fi and verify the Firmware URL on the Setup page.
OTA shows ERROR2 / ERROR3	Firmware server unreachable or version file invalid	Check internet access; restore the Firmware URL to its default if it has been changed.
OTA shows ERROR4	Download or flash write failed	Power-cycle and try again. If it persists, perform a manual reflash via USB.
Live web page will not load	Live Form disabled or wrong password	Verify Live Form is Enabled in AP setup, and the Live Form Password matches.
Backlight LEDs not lighting	Brightness set to 0, or LEDs disabled by override	Check Backlight Brightness; verify no period override is forcing them off.
Display dims unexpectedly at night	Period override active	Press SW1 to wake the display temporarily, or adjust the override on the Overrides page.

Copyright and Disclaimer

Nixie variants of this clock contain a small amount of mercury and neon gas under low pressure and operate from an internal high-voltage supply (~170 V). Operate the clock at the supplied voltage only; do not open or service the high-voltage circuitry.

A good-faith effort has been made to remove offensive content based on race, religion, or sexual orientation from the included libraries. Please report any remaining offensive entries, together with the library name and the specific word or phrase, to the distributor.

WordClock — Designed by Mitchell Feig